Aryan Bokolia

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LinkedIn | GitHub | Location: Mangaluru, India | **CGPA: 7.43**

Education

National Institute of Technology Karnataka (NITK), Surathkal — **Aug 2024 – Present**
B.Tech in Computer Science and Engineering — *Mangaluru, India*

- Focusing on foundational excellence in Data Structures & Algorithms, Digital Logic, and Linear Algebra.

Technical Skills

Programming Languages: C, C++, Python, JavaScript, HTML5, CSS3, SQL (MySQL)
Web Development & Frameworks: React.js, Next.js, Node.js, Express.js, Vite, Tailwind CSS, Full Stack Development
Systems & Networking: eBPF/XDP, BPF Maps, Linux Kernel Programming, Mininet, libbpf, Network Slicing
Database & Backend: Firebase, MySQL, SQLite, WebSockets, REST APIs
Tools & Technologies: Git, GitHub, Docker, Kali Linux, Wireshark, VS Code, Logisim, Verilog
Core Competencies: Data Structures, Algorithms, Cybersecurity, Network Security, Problem Solving, Team Leadership

Projects

Traffic Correlation Research Platform | ***Python, Node.js, Express.js, Next.js, SQLite, WebSockets*** — GitHub

- Engineered a full-stack cybersecurity research platform for traffic correlation analysis on anonymous networks (Tor), supporting 4 capture modes: simulation, replay, PCAP, and live capture.
- Implemented a Python analysis engine with feature extraction, probabilistic correlation scoring, and region estimation, achieving end-to-end analysis across 100+ session records.
- Built an interactive Next.js dashboard visualizing inferred network paths, protocol summaries, and confidence-based analysis, enabling real-time session monitoring for 5+ concurrent research workflows.

AeroNITK Club Website | ***React.js, Vite, Tailwind CSS*** — GitHub

- Independently designed and deployed the club's first official website from scratch, increasing online visibility and member engagement by 3x across 200+ active members.
- Architected 10+ modular sections including events, projects, and team showcases, reducing information retrieval time for members by 60%.
- Built a fully responsive UI with optimized asset loading, reducing bounce rate by 30% and ensuring seamless cross-device experience.

eBPF-XDP Network Slicing Firewall | ***C (eBPF/XDP), Python, libbpf, Mininet, BPF Maps*** — GitHub

- Implemented a kernel-level XDP packet-processing firewall based on IEEE 6G network slicing research, classifying IPv4, TCP, UDP, ICMP, VXLAN, and GTP-U traffic into application-specific slices (PASS, DROP, STEER) at wire speed with sub-millisecond decision latency.
- Designed 9 BPF maps for high-speed packet decisions including LPM trie CIDR blocking with 1,636 Spamhaus threat-intelligence ranges, enforcing MAC, IP, port, and rate-limit firewall rules entirely in kernel space.

Experience & Leadership

Indian Society for Technical Education (ISTE) NITK — **Sep 2025 – Present**
Crypt SIG Coordinator (Apr 2026 – Present) | Executive Member – Technical Team — *Mangaluru, India*

- Lead the Cryptography Special Interest Group, organizing cybersecurity workshops, CTF training sessions, and technical seminars for 200+ student members.
- Coordinated technical initiatives and events fostering skill development in cryptography, network security, and emerging technologies across the student body.

Team AeroNITK — **Apr 2025 – Present**
Web Dev Lead (Apr 2026 – Present) | Web Team Member — *Mangaluru, India*

- Solely built and deployed the complete AeroNITK official website, serving 200+ club members, and currently leading the web development team as Web Dev Lead.
- Mentored junior members in React.js, modern frontend best practices, and responsive design, improving team delivery speed by 2x.

Achievements & Certifications

Competitive Achievement: Secured **1st Place** in WEC NITK Capture The Flag (CTF), an open cybersecurity competition with participation from 20+ diverse teams
Research Implementation: Implemented IEEE-published eBPF-XDP network slicing architecture (6G front-haul research) as a functional kernel-level firewall prototype